

S. VIKAS

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PROFESSIONAL SUMMARY

Performance-driven Full-Stack Developer with expertise in building scalable, production-grade web applications utilizing **React.js**, **Node.js/Express.js**, and **PostgreSQL**. Proven track record in designing dual-database architectures, automated eligibility engines, secure E2E encryption systems, and real-time event portals. National-level hackathon winner recognized for technical leadership and writing **clean, modular, and highly optimized code** with robust CI/CD integration (GitHub Actions, Vercel). Experienced in managing dynamic timelines, role-based workflows, and complex asynchronous state.

TECHNICAL SKILLS

Programming & Frameworks: * **Languages:** JavaScript (ES6+), TypeScript, SQL, HTML5/CSS3, Python, Java, C/C++, PHP * **Frontend:** React.js (Hooks: useState/useEffect, Props, State Management, Performance Optimization), React Router, Next.js, Tailwind CSS v4, Framer Motion, GSAP * **Backend:** Node.js (Event Loop, Async/Await runtime performance), Express.js (Middleware, Error Handling), RESTful API Design (HTTP Methods, Status Codes), Next.js App Router, Spring Boot * **Databases:** PostgreSQL (Schema Design, Joins, Indexing, Transactions), MySQL, MongoDB, Firebase, FAISS **Data & Infrastructure:** * **Cloud & DevOps:** Vercel (Hosting), Render, GitHub Actions (CI/CD Pipelines), Docker, Environment Variables (Secrets Management) * **Testing & Tools:** Git (Branching, PRs), Postman (API Testing), Jest (Testing/Mocking), VS Code, Chrome DevTools (Debugging) * **Core Concepts:** Data Structures & Algorithms (DSA), System Design, Object-Oriented Programming (OOPS), Asynchronous Programming, Agile Methodologies

PROFESSIONAL EXPERIENCE

Web Development Intern | *Office of Dean Academics, Veltech University*
| Aug 2025 – Present * **Campus Recruitment Portal (C2C):** Architected a centralized PHP/MySQL placement portal featuring a dual-database design (MSSQL/MySQL) to securely sync academic records. Developed an automated eligibility engine and integrated a custom ATS using PDF parsing to score student resumes. (Link) * **Research Studies Management System:** Engineered a role-based portal to manage PhD scholar progress, digitizing the document review

pipeline. Built an automated, dynamic timeline tracking engine to calculate stage deadlines and handle complex fractional month logic. (Link) * **R&D Management System:** Architected a data-driven dashboard for the Office of Research and Development to track institutional metrics. (Link) * **E-Content Platform (VE_LEARN):** Built an educational content delivery app focusing on user experience and accessibility. (Link)

AI Engineer Intern | *Siter Academy* | Aug 2025 – Oct 2025 * **AI Pipelines:** Developed production-level AI product pipelines and recommendation engines emphasizing scalability and modularity. * **Engineering Quality:** Participated in Agile workflows, code reviews, and sprint planning, ensuring high code quality and test coverage. * **Data Modeling:** Optimized feature engineering and data preparation workflows for ML model training and evaluation.

Project Intern | *Ashok Leyland (Guided by DGM)* | Mar 2024 – Dec 2024 * **Agri-Bot Development:** Owned the full lifecycle of an autonomous agricultural robot from conceptual design to working hardware prototype. * **R&D Engineering:** Applied analytical methodologies to solve real-world automation challenges; work contributed to **patent filings and technical research publications**.

KEY PROJECTS

Tantraz 2026 – Technical Symposium Portal

- **Tech Stack:** Next.js (React 19, App Router), TypeScript, Tailwind CSS v4, Framer Motion, GSAP, NextAuth.js, Google Sheets API v4, PostgreSQL, Zod
- Designed an immersive web portal with custom GSAP text-reveal, Framer Motion neon glitches, and canvas particle backgrounds.
- Built a serverless registration endpoint with Zod schema-based request validation, syncing attendee data to a Google Sheets API v4 backend.
- Configured **PostgreSQL** database schemas for storing registration logs and integrated secure credentials-based auth with **NextAuth.js** to lock administrative routes.
- **Link:** <https://tantraz-2026.vercel.app>

E2EE Chat System – Secure Instant Messaging Platform

- **Tech Stack:** React (Vite, TypeScript), Node.js, Express.js, Socket.io, WebCrypto API (RSA, AES), PostgreSQL, JWT
- Engineered a **full-stack end-to-end encrypted (E2EE)** chat application ensuring zero-knowledge privacy; backend handles routing without reading payloads.
- Implemented client-side cryptography using **WebCrypto API** for **RSA key generation**, key exchanges, and AES-CBC symmetric encryption.

- Architected real-time messaging using **Socket.io** on a Node.js/Express backend with custom middleware for JWT validation and PostgreSQL transaction state tracking.
- Utilized React state management (**Hooks: useState, useEffect**) and context APIs to handle asynchronous WebSocket lifecycles and dynamically render chat histories.

Farmer AI Agent – Intelligent Multilingual Decision Support System

- **Tech Stack:** Python, LangChain, React, FAISS, Whisper STT, Flask
- Built a **RAG-based AI assistant** providing real-time crop advisory via voice and text in multiple languages.
- Architected low-latency data retrieval using **vector search** and optimized the pipeline for deployment in low-connectivity environments.

ISL Multilingual Translator – AI-Powered Accessibility Engine

- **Tech Stack:** TensorFlow, OpenCV, React, Flask
- Developed a real-time **Indian Sign Language (ISL) to speech** translator to aid communication for the hearing-impaired.
- Designed a high-throughput inference layer server-side with a responsive, component-based React frontend.

Autonomous Agricultural Robot – Precision Automation & Vision

- **Tech Stack:** C++, Python, YOLOv8, RKNN, Rock 5 ITX, Teensy 4.1
- Engineered an on-device automation system for weed detection and removal using real-time **Computer Vision**.
- Synthesized embedded control logic with deep learning inference, managing high-frequency sensor streams (LIDAR, Encoders).

MedPal – AI Healthcare Symptom Analyzer

- **Tech Stack:** Python, NLP, LangChain, Vector Databases
- Designed a modular symptom-analysis engine that offers structured, context-aware medical guidance.
- Implemented intelligent knowledge lookup via vector retrieval for medical diagnostic support.

EDUCATION

B. Tech in Computer Science and Engineering | CGPA: 8.58/10

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai | 2023 – 2027

* **Core Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning, Computer Networks, Software Engineering.

HONORS & ACHIEVEMENTS

- **Winner** – Agentic AI Hackathon 2025 (World’s First Prompt-Exclusive Hackathon, SIMATS)
 - **Performance Optimization:** Highly proficient in optimizing asynchronous JS/Node.js runtimes (Event Loop, Non-blocking I/O) and memory-profiling React applications to eliminate rendering bottlenecks.
 - **National Rank 37/150** – TechXcelerate Hackathon, BITS Goa
 - **Top 5%** – ODOO X MSU FootPrints’25 Hackathon
 - **First Prize** – Paper Presentation Competition, ISTE Tamil Nadu (PSR College)
 - **Qualifier** – Smart India Hackathon (SIH) Internal Rounds (2024, 2025)
 - **Competitive Programming:** Active solver on platforms focusing on Trees, Graphs, DP, and Bit Manipulation.
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PATENTS & PUBLICATIONS

- **Patent Filed:** Automatic Railway Gate System (2023)
- **Research (Under Review):** Real-Time Autonomous Navigation for Ag-Bots
- **Research (Under Review):** Indian Sign Language Recognition using Deep Learning
- **Patent:** Currently filing for the Ashok Leyland Agri-Bot project (2024)